IPE - Math - 2 classes

ABC Analysis Example

Part Number	Description	Annual Demand	Unit Value ($)	Annual Consumption ($)
1	Boxes	500	3	1500
2	Cardboard	18000	0.02	360
3	Cover Stock	10000	0.75	7500
4	Glue	75	40	3000
5	Inside Cover	20000	0.05	1000
6	Reinforcing Tape	3000	0.15	450
7	Ink	150000	0.45	67500

	Part Number	Description	Annual Demand	Unit Value ($)	Annual Consumption ($)	Annual Demand (%)	Annual Consumption (%)
83%	7	Ink	150000	0.45	67500	74.41	83.02
13%	3	Cover Stock	10000	0.75	7500	4.96	9.22
	4	Glue	75	40	3000	0.04	3.69
4%	1	Boxes	500	3	1500	0.25	1.84
	5	Insite Cover	20000	0.05	1000	9.92	1.23
	6	Reinforcing Tape	3000	0.25	950	1.49	0.55
	2	Cardboard	18000	0.02	360	8.93	0.44
	Totals		201575		81320		

IM - Math

Weighted Moving Average

Exponential Smoothing

New forecast = Last period's forecast + α (Last period's actual demand - Last period's forecast)

$$F_t = F_{t-1} + \alpha (A_{t-1} - F_{t-1})$$

F_t = new forecast

F_{t-1} = previous forecast

α = smoothing (or weighting) constant ($0 \leq \alpha \leq 1$)

A_{t-1} = ~~last~~ previous actual demand

New forecast = $142 + 0.2(153 - 142)$

$= 144.2$

≈ 144 cars

Forecast error = actual demand - fore

Mean Absolute Deviation (MAD)

$$MAD = \frac{\sum |\text{Actual} - \text{Forecast}|}{n}$$

Mean Squared Error (MSE)

$$MSE = \frac{\sum (\text{Forecast error})^2}{n}$$

Mean Absolute Percent Error (MAPE)

$$MAPE = \frac{\sum_{i=1}^{n} 100 |\text{Actual}_i - \text{Forecast}_i| / \text{Actual}_i}{n}$$

Q)

Quarter	Actual	Forecast with $\alpha = 0.10$	Absolute Deviation for $\alpha = 0.10$	Forecast with $\alpha = 0.50$	Absolute Deviation for $\alpha = 0.50$
1	180	175	5	175	5
2	168	175.5	7.5	177.5	9.5
3	159	174.75	15.75	172.75	13.75
4	175	173.18	1.82	165.88	9.12
5	190	173.36	16.64	170.44	19.56
6	205	175.02	29.98	180.22	24.78
7	180	178.02	1.98	192.61	12.61
8	~~180~~ 182	178.22	3.78	186.31	4.31
			82.45		~~[illegible]~~ 98.63

$$MAD = \frac{\Sigma|Actual - Forecast|}{n}$$

$$= \frac{82.45}{8} \quad [\text{For } \alpha = 0.1]$$

$$= 10.30$$

$$MAD = \frac{98.63}{8} \quad [\text{For } \alpha = 0.5]$$

$$= 12.33$$

$$MSE = \frac{\Sigma(\text{Forecast error})^2}{n}$$

$$\cancel{= \frac{(82.45)^2}{8} \quad [\text{For } \alpha = 0.1]}$$

$$= \frac{1526.5237}{8} \quad [\text{For } \alpha = 0.1]$$

$$= 190.82$$

$$\cancel{MSE = \frac{(98.63)^2}{8} \quad [\text{For } \alpha = 0.5]}$$

$$= \frac{\quad}{8} \quad [\text{For } \alpha = 0.5]$$

$$MAPE = \frac{\Sigma\, 100|\text{deviation}|/\text{Actual}}{n}$$

$$= \cancel{\frac{25}{19} + \frac{125}{28} + \frac{525}{52}} + \frac{44.7469}{8}$$

[For $\alpha = 0.1$]

$$= 5.59\%$$

$$MAPE = \frac{1}{8} \quad [\text{For } \alpha = 0.5]$$

IPE - Math

MAD = 10.30 → The range of deviation between actual and forecasted value

MSE = 190.81 → যত বেশি তত বেশি error

MAPE = 5.59% → যত " " " "